

A chatbot acts like a typical software application. It has an app layer, a database (knowledge base) and APIs to call other external administrators. Most of the bots are trained using past information so organisations have discussion logs to upgrade the chatbot's capability. Developers use these logs to analyse what clients are trying to ask. With a blend of machine learning tools and models, developers coordinate client inquiries and reply with the most appropriate answer. Figure 2 highlights how a chatbot works.

Architecture and workflow models of chatbots

Listed below are the three ways in which chatbots work.

Pattern matching: This is used to perform text grouping and to produce the desired response to the clients. In this model, Artificial Intelligence Markup Language (AIML) is used to model the patterns. As an example, an end user may ask the chatbot, "What is the capital of India?" And the bot will answer, "According to Google/Wikipedia, New Delhi is the capital of India."

For every sort of question, a pattern must be accessible in the database to give a reasonable response. With a number of pattern combinations, the bot makes a hierarchical structure. Various pattern matching algorithms are used to classify and produce the most efficient structure.

Natural language understanding (NLU): In this method, the following help the chatbot's functionality.

Entities: The type of chatbot and how it works in real-time, e.g, a chatbot that handles technical support for laptop customers.

Context: When a natural language understanding algorithm examines a sentence, it doesn't have the historical backdrop of the user's text conversation. This implies that, if it gets a response to a question it has been recently asked, it won't recall the inquiry. So, the conversations in this method are stored under separate categories, such as "Ordering Pizza."

Expectations: A chatbot should answer all the queries of the end users, for instance, during an online shopping interaction.

Natural Language Processing (NLP): This converts the user's speech or text into efficient structured data, and then the analysis of the data is used to provide the relevant answers. This includes the following steps.

Tokenisation: The NLP separates a series of words into tokens or pieces that are linguistically representable, with a different value in the application.

Sentiment analysis: It will study and learn the user's experiences, and transfer the inquiry to a human when necessary.

Normalisation: This program model processes the text to find out the typographical errors and common spelling mistakes that might alter the intended meaning of the user's request.

Named entity recognition: The program model of a chatbot looks for different categories of words, similar to the name of the particular product, the user's address or name, or whatever information is required.

Dependency parsing: The chatbot searches for the subjects, verbs, objects, common phrases and nouns in the user's text to discover the related phrases that the users want to convey.

The advantages of chatbots

The following benefits can be derived from chatbots in real-world applications.

- **Availability:** Chatbots, being AI driven software, are operational and ready to engage with end users 24x7, 365 days a year, with instant replies, and can handle a high volume of tasks.
- **Increased efficiency:** With chatbots, faster service can be delivered by eliminating all sorts of manual processes. Besides, with automation, overall costs have decreased. Personnel can now work on strategic business functions, which helps to control overall labour costs.

- **Better customer support:** With chatbots, businesses get better clarity about their customers' requirements, as the bots provide better answers to users. With the passage of time, using machine learning and deep learning techniques, chatbots become intelligent by gaining an in-depth knowledge about customers.
- **Platform flexibility:** Chatbots can be designed on varied platforms, and easily adapt and integrate with dynamic business operations to meet customer requirements in whatever platforms they are native to.
- **Endless support and strong programmability:** Bots are as yet incapable of handling customers who sometimes lose their patience. But since they are on digital platforms, where people now spend a majority of their waking hours working, they can be best used to automate common tasks such as arranging meetings, providing advanced search functionality, and so on.

Real-time chatbot applications

Table 1 highlights the major use cases of real-time chatbot applications.

Open source chatbot platforms

A chatbot platform or chatbot software is a computer program designed to simulate an intelligent conversation with one or more humans using auditory or textual methods. It is programmed with artificial intelligence (AI) that enables it to conduct a conversation in the same way as a human would. Chatbots are designed to handle enquiries by responding in the same way as a person would. The most effective and convincing systems have complex natural language processors (NLP), which may even display a human face and be able to understand almost any form of input, and respond appropriately.

A chatbot development platform is a tool using which one can create a chatbot. The number of chatbot-related